



Day 4 Task - Evidence Hashing & Integrity Verification



Trios cyber • 6 Sept (Edited 9 Sept)

100/100 | Due 7 Sept, 23:40

Practical Assignment: Evidence Hashing & Integrity Verification

Dear Students,

For this practical assignment, you are required to calculate and compare hash values for digital evidence and verify the integrity of an acquired forensic image.

The objective of this exercise is to understand how hashing is used to verify that acquired evidence corresponds to the evidence that was collected.

Part 1: Understand Hash Values

Study the concept of cryptographic hashing and its use in digital forensics.

For this practical, work with the forensic image created in Task 1. If it's not available, please create a new one.

Record the available hash information associated with your forensic image.

Part 2: Calculate/Obtain Hash Values

Using FTK Imager or another approved hashing utility:

1. Identify the file or forensic image to be examined.
2. Obtain its available hash value(s).
3. Record the hash algorithm used.
4. Record the complete hash value.
5. Take a screenshot showing the hash calculation/result.

Where available, examine both MD5 and SHA-256 values.

Part 3: Verify the Forensic Image

Use the verification function available in FTK Imager for the forensic image, where applicable.

Document:



- Image/file being verified
- Hash algorithm
- Calculated hash
- Verification/reference hash
- Whether the values match
- Verification result

Take a clear screenshot of the verification result.

Part 4: Demonstrate the Effect of a Change

Create a copy of a small test file and calculate its hash.

Make a minor change to the test copy, such as adding or removing a small amount of text, and calculate its hash again.

Compare the original and modified hash values.

Do not modify the forensic image or original evidence created in Task 1.

Document your observation and explain why even a small change can produce a different hash value.

Part 5: Evidence Integrity

Answer the following questions:

1. What is a hash value?
2. Why is hashing important in digital forensics?
3. What does a matching hash value indicate?
4. Why should the original evidence not be modified?
5. How do hash values support evidence integrity and verification?

Mention the answers to these five questions specifically in your report.

Part 6: Document Your Work

Prepare a PDF or DOCX report containing:

1. Student Information
2. Objective
3. Hashing procedure
4. Hash algorithm(s) used
5. Hash values obtained
6. Verification result
7. Original vs modified test-file comparison
8. Screenshots
9. Answers to the evidence-integrity questions
10. Conclusion



Each screenshot should have a short caption explaining what it demonstrates.

Submission Requirements

Submit ONE file only:

- PDF (.pdf) OR
- Microsoft Word document (.docx)

The report must contain both written steps and screenshots/evidence of the completed work.

Screenshots alone will not be considered a complete submission.

Important Notes

Do not modify the original forensic image or original evidence.

Perform the modification experiment only on a separate test copy.

Record complete hash values accurately.

Expected Learning Outcome

After completing this practical, you should be able to:

1. Explain the purpose of hashing in digital forensics.
2. Obtain and record hash values.
3. Verify the integrity of a forensic image.
4. Compare hash values before and after a change.
5. Explain how hashing supports forensic evidence integrity.

Submission: Upload your completed PDF/DOCX report to this Google Classroom assignment before the specified deadline. Late submissions are allowed only till after 48 hours from the time the task was posted but it may result in deduction of marks.

Please read the instructions carefully before beginning the assignment.

Your work

Marked

Day 4.pdf
PDF



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